Imagine that unperturbed problem involves <u>degenerate</u> states (happens all the time: excited state of hydrogen atom (linear Stark effect), 2D and 3D harmonic oscillator; etc).

That, $\hat{H}_0 |\varphi^{(0)}_{n,i}\rangle = E_n^{(0)} |\varphi^{(0)}_{n,i}\rangle$ where $i = 1, 2 \ldots N$

runs over degenerate states, all with the same energy $E_n^{(0)}$ [$\underline{i}$-independent!!!]

Then, e.g., $$E_n^{(2)} = \sum_{k \neq n}{}' \frac{|\langle \varphi_k^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

would involve terms such as [because $k = (n,2) \neq n = (n,1)$]

$$\frac{\langle \varphi^{(0)}_{n,2} | \hat{H}_1 | \varphi^{(0)}_{n,1} \rangle}{E_n^{(0)} - E_n^{(0)}} = \infty \qquad \boxed{\text{unless } \langle \varphi^{(0)}_{n,2} | \hat{H}_1 | \varphi^{(0)}_{n,1} \rangle = 0 \;!!!}$$

$\Rightarrow$ have to re-analyze equations for this important case.

Imagine $N=2$: $\Rightarrow$ there are 2 states $|\varphi^{(0)}_{n,1}\rangle$ and $|\varphi^{(0)}_{n,2}\rangle$

Note that any linear combination of them

$$|\gamma_1\rangle = \alpha |\varphi^{(0)}_{n,1}\rangle + \beta |\varphi^{(0)}_{n,2}\rangle \qquad \hat{H}_0 |\gamma_1\rangle = E_n^{(0)} |\gamma_1\rangle$$

$$\langle \gamma_1 | \gamma_1 \rangle = |\alpha|^2 + |\beta|^2 = 1 \Rightarrow \text{choose } \beta = \sqrt{1-\alpha^2} \ \underline{\text{real}}$$

Then another (<u>orthogonal</u>) combination is

$$|\gamma_2\rangle = -\beta |\varphi^{(0)}_{n,1}\rangle + \alpha |\varphi^{(0)}_{n,2}\rangle$$

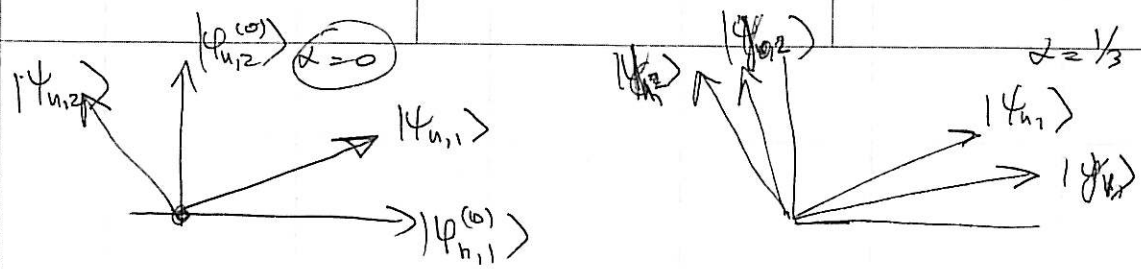
Indeed

$$\langle \gamma_1 | \gamma_2 \rangle = \left(\langle \varphi^{(0)}_{n,1} | \alpha^* + \langle \varphi^{(0)}_{n,2} | \beta^* \right) \left(-\beta |\varphi^{(0)}_{n,1}\rangle + \alpha |\varphi^{(0)}_{n,2}\rangle \right)$$
$$= -\alpha^* \beta + \beta^* \alpha = 0 \quad (\text{for } \alpha, \beta = \text{real}).$$

Thus there is an <u>infinite</u> choice of possible <u>degenerate</u> states

$$\begin{cases} |\gamma_1\rangle = \alpha |\varphi^{(0)}_{n,1}\rangle + \sqrt{1-\alpha^2} |\varphi^{(0)}_{n,2}\rangle \\ |\gamma_2\rangle = -\sqrt{1-\alpha^2} |\varphi^{(0)}_{n,1}\rangle + \alpha |\varphi^{(0)}_{n,2}\rangle \end{cases}$$

with $\alpha \in [0,1]$



~~Expect~~ two exact vectors $|\psi_{n,1,2}\rangle$: some particular combination $(|\gamma_1\rangle, |\gamma_2\rangle)$ may be sufficiently <u>close to</u> the exact states $(|\psi_{n1}\rangle, |\psi_{n2}\rangle)$ and provides the best starting point.

let's find it:

$$|\psi_n\rangle = \sum_{i=1}^{N} c_i \, |\varphi^{(0)}_{n,i}\rangle + \lambda \, |\varphi^{(1)}_n\rangle + \dots$$

where $\{c_i\}$ are not known yet [ours α, and β]

From $(\hat{H}_0 + \hat{H}_1) \, |\psi_n\rangle = (E_n^{(0)} + E_n^{(1)}) \, |\psi_n\rangle$

find at λ^1-order

$$\hat{H}_0 \, |\varphi^{(1)}_n\rangle + \hat{H}_1 \sum_{i=1}^{N} c_i \, |\varphi^{(0)}_{ni}\rangle = E_n^{(0)} \, |\varphi^{(1)}_n\rangle + E_n^{(1)} \sum_{i=1}^{N} c_i \, |\varphi^{(0)}_{ni}\rangle$$

Multiply by $\langle \varphi^{(0)}_{nj}|$ $(j = 1, \dots N)$:

$$\underbrace{\langle \varphi^{(0)}_{nj} | \hat{H}_0 | \varphi^{(1)}_{ni} \rangle}_{E_n^{(0)} \delta_{ij}} + \sum_{i=1}^{N} c_i \langle \varphi^{(0)}_{nj} | \hat{H}_1 | \varphi^{(0)}_{ni} \rangle = E_n^{(0)} \delta_{ij} + E_n^{(1)} \sum_{i=1}^{N} c_i \underbrace{\langle \varphi^{(0)}_{nj} | \varphi^{(0)}_{ni} \rangle}_{\delta_{ij}}$$

$$\Rightarrow \sum_{i=1}^{N} \langle \varphi^{(0)}_{nj} | \hat{H}_1 | \varphi^{(0)}_{ni} \rangle \, c_i = E_n^{(1)} \, c_j$$

true for any $j = (1, \dots N)$ $\Rightarrow$ matrix problem

$$\begin{pmatrix} & & \\ & (\hat{H}_1)_{ji} & \\ & & \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} = E_n^{(1)} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

$$(\hat{H}_1)_{ji} = \langle \varphi^{(0)}_{nj} | \hat{H}_1 | \varphi^{(0)}_{ni} \rangle$$

For $N=2$, for example:

$$\begin{cases} \langle \varphi_{n1}^{(0)} | \hat{H}_1 | \varphi_{n1}^{(0)} \rangle c_1 + \langle \varphi_{n1}^{(0)} | \hat{H}_1 | \varphi_{n2}^{(0)} \rangle c_2 = E_n^{(1)} c_1 \\ \langle \varphi_{n2}^{(0)} | \hat{H}_1 | \varphi_{n1}^{(0)} \rangle c_1 + \langle \varphi_{n2}^{(0)} | \hat{H}_1 | \varphi_{n2}^{(0)} \rangle c_2 = E_n^{(1)} c_2 \end{cases}$$

$$\Rightarrow \begin{pmatrix} (H_1)_{11} & (H_1)_{12} \\ (H_1)_{21} & (H_1)_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_n^{(1)} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Thus the problem is reduced to the ~~diag~~ specific eigenvalue problem: we diagonalize the perturbing $\hat{H}_1$ in the <u>subspace of degenerate states of $\hat{H}_0$</u>

Ex in the book: Stark effect

first-order correction to $E_n^{(1)}$ (there will be 2 of them) is the <u>eigenvalue</u>; the "right" combination $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ is the <u>eigenstate</u>.

Example Consider 2D harmonic oscillator:

$$\hat{H}_0 = \frac{\hat{p}_x^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 + \frac{\hat{p}_y^2}{2m} + \frac{1}{2} m\omega^2 \hat{y}^2$$

Clearly $\hat{H}_0 = \hbar\omega\left(\hat{a}_x^+ a_x + \frac{1}{2}\right) + \hbar\omega\left(\hat{a}_y^+ a_y + \frac{1}{2}\right)$

and the eigenstates are: $\Psi_{n_1 n_2}(x,y) = \Psi_{n_1}(x)\cdot\Psi_{n_2}(y)$
(recall separation of variables!)

$\Rightarrow$ eigenstates $E^{(0)}_{n_1 n_2} = E^{(0)}_N = \hbar\omega\left(n_1 + \frac{1}{2} + n_2 + \frac{1}{2}\right) = \hbar\omega(N+1)$
$N = n_1 + n_2 = 0, 1, 2 \ldots$

$\Rightarrow$ (A) the ground state: $n_1 = 0, n_2 = 0$, $N = 0$ $\Rightarrow$ non-degenerate
$E^{(0)}_0 = \hbar\omega$

(B) 1st excited state: $N = 1$, $E^{(0)}_1 = 2\hbar\omega$ $\Rightarrow$ 2-fold degenerate

$N = 1 + 0$ $\quad n_1 = 1, n_2 = 0$
$N = 0 + 1$ $\quad n_1 = 0 \quad n_2 = 1$

(C) 2nd excited state: $N = 2$, $E^{(0)}_2 = 3\hbar\omega$ $\Rightarrow$ 3-fold degenerate

	n_1	n_2
$N = 2 + 0$	2	0
$N = 0 + 2$	0	2
$N = 1 + 1$	1	1

$\longrightarrow$ Now we apply perturbation $\boxed{\hat{H}_1 = \alpha\hat{x}\hat{y}}$ (such that separation of variables does not work)

$E^{(1)}_0$ and $E^{(2)}_0$

① Find leading correction to the ground state energy of $\hat{H}_0$.

$E^{(0)}_0 = \hbar\omega \qquad \Psi^{(0)}_{00}(x\,y) = \Psi_0(x)\,\Psi_0(y)$

$$\Rightarrow E^{(1)}_0 = \langle \Psi^{(0)}_{00} | \hat{H}_1 | \Psi^{(0)}_{00} \rangle = \alpha \langle \Psi_0(x) | \hat{x} | \Psi_0(x) \rangle \langle \Psi_0(y) | \hat{y} | \Psi_0(y) \rangle = 0.$$

$\Rightarrow$ consider $$E^{(2)}_0 = \sum_{k \neq 0}{}' \frac{|\langle \Psi^{(0)}_k | \hat{H}_1 | \Psi^{(0)}_{00} \rangle|^2}{E^{(0)}_k - \hbar\omega}$$

Clearly $\underline{k}$ becomes a set of two indices now:
$k = (n_1, n_2)$

$$\Rightarrow E^{(2)}_0 = -\sum_{n_1 n_2 \neq 0}{}' \frac{|\langle \Psi^{(0)}_{n_1 n_2} | \hat{H}_1 | \Psi^{(0)}_{00} \rangle|^2}{E^{(0)}_{n_1 n_2} - \hbar\omega} = -\alpha^2 \sum_{n_1 \neq 0}{}' \sum_{n_2 \neq 0}{}' |\langle \Psi^{(0)}_{n_1} | \hat{x} | \Psi^{(0)}_0 \rangle|^2 \cdot |\langle \Psi^{(0)}_{n_2} | \hat{y} | \Psi^{(0)}_0 \rangle|^2 \frac{1}{\hbar\omega(n_1 + n_2 + 1) - \hbar\omega}$$

Hence

$$E_0^{(2)} = \frac{-\alpha^2}{\hbar\omega} \sum_{n_1 n_2}{}' \frac{|\langle \varphi_{n_1}^{(0)} | \hat{x} | \varphi_0^{(0)} \rangle|^2 \, |\langle \varphi_{n_2}^{(0)} | \hat{y} | \varphi_0^{(0)} \rangle|^2}{n_1 + n_2}$$

but $\langle \varphi_{n_1}^{(0)} | \hat{x} | \varphi_0^{(0)} \rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle n_1 | \hat{a}_x + \hat{a}_x^+ | 0 \rangle = \sqrt{\frac{\hbar}{2m\omega}} \, \delta_{n_1,1}$

$$\Rightarrow E_0^{(2)} = \frac{-\alpha^2}{\hbar\omega} \sum_{\substack{n_1=1 \\ n_2=1}} \left(\frac{\hbar}{2m\omega}\right)^{1/2 \cdot 2 \cdot 2} \frac{1}{n_1+n_2} =$$

$$= \frac{-\alpha^2}{\hbar\omega} \left(\frac{\hbar}{2m\omega}\right)^2 \frac{1}{2} = -\frac{\alpha^2 \hbar}{8 m^2 \omega^3}$$

$[\alpha] = m\omega^2 \Rightarrow [E_0^{(2)}] = \frac{m^2 \omega^4 \hbar}{m^2 \omega^3} = \hbar\omega$ OK.

NOTE

$\cancel{\frac{1}{2} m\omega^2 \left(x^2 + y^2 + \frac{2\alpha}{m\omega^2} xy\right) = P.E.}$

Exact solution

$$P.E. = \frac{1}{2} m\omega^2 (x^2 + y^2) + \alpha xy$$

Introduce $\left.\begin{aligned} x_1 &= \frac{1}{\sqrt{2}}(x+y) \\ y_1 &= \frac{1}{\sqrt{2}}(-x+y) \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} x &= \frac{1}{\sqrt{2}}(x_1 - y_1) \\ y &= \frac{1}{\sqrt{2}}(x_1 + y_1) \end{aligned}\right.$

$$\Rightarrow \frac{1}{2} m\omega^2 (x^2+y^2) = \frac{1}{2} m\omega^2 \cdot \frac{1}{2}\left[(x_1 - y_1)^2 + (x_1+y_1)^2\right] = \frac{1}{2} m\omega^2 (x_1^2 + y_1^2)$$

but $\alpha xy = \frac{\alpha}{2}(x_1^2 - y_1^2)$

$$\Rightarrow P.E. = \frac{1}{2}(m\omega^2 + \alpha)\, x_1^2 + \frac{1}{2}(m\omega^2 - \alpha)\, y_1^2$$

Also $\frac{\partial}{\partial x} = \frac{\partial}{\partial x_1}\frac{\partial x_1}{\partial x} + \frac{\partial}{\partial y_1}\frac{\partial y_1}{\partial x} = \frac{1}{\sqrt{2}}\left(\frac{\partial}{\partial x_1} - \frac{\partial}{\partial y_1}\right)$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial x_1}\frac{\partial x_1}{\partial y} + \frac{\partial}{\partial y_1}\frac{\partial y_1}{\partial y} = \frac{1}{\sqrt{2}}\left(\frac{\partial}{\partial x_1} + \frac{\partial}{\partial y_1}\right)$$

$\Rightarrow \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial y_1^2} \Rightarrow$ K.E. is invariant as well.

Thus $\hat{H}_0 \to -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_1^2} + \frac{1}{2}(m\omega^2 + \alpha)\, x_1^2 - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial y_1^2} + \frac{1}{2}(m\omega^2 - \alpha)\, y_1^2$

$\Rightarrow$ two oscillators with frequencies: $\frac{1}{2} m\Omega_1^2 = \frac{1}{2}(m\omega^2 + \alpha)$

$\frac{1}{2} m\Omega_2^2 = \frac{1}{2}(m\omega^2 - \alpha)$

$$\Rightarrow \Omega_{1,2} = \sqrt{\omega^2 \pm \frac{\alpha}{m}}$$

(6.37)

$$\Rightarrow E = \hbar\Omega_1\left(\nu_1+\tfrac{1}{2}\right) + \hbar\Omega_2\left(\nu_2+\tfrac{1}{2}\right)$$

$$\Rightarrow \text{G.S.} \quad E_{g.s.} = \tfrac{1}{2}\hbar(\Omega_1+\Omega_2) \quad (\nu_1=\nu_2=0)$$

$$E_{g.s.} = \tfrac{1}{2}\hbar\left(\sqrt{\omega^2+\frac{\alpha}{m}} + \sqrt{\omega^2-\frac{\alpha}{m}}\right) = \tfrac{1}{2}\hbar\omega\left(\sqrt{1+\frac{\alpha}{m\omega^2}} + \sqrt{1-\frac{\alpha}{m\omega^2}}\right)$$

$$= \tfrac{1}{2}\hbar\omega\left[1+\tfrac{1}{2}\frac{\alpha}{m\omega^2} - \tfrac{1}{8}\left(\frac{\alpha}{m\omega^2}\right)^2 + 1 - \tfrac{1}{2}\frac{\alpha}{m\omega^2} - \tfrac{1}{8}\left(\frac{\alpha}{m\omega^2}\right)^2\right]$$

$$= \tfrac{1}{2}\hbar\omega\left(2 - \tfrac{1}{4}\frac{\alpha^2}{m^2\omega^4}\right) = \hbar\omega - \frac{\alpha^2\hbar}{8m^2\omega^3} \quad \text{(ok)}.$$

---

(2) Now find corrections to the 1st excited state! $n_1=1 \; n_2=0$

$E_1^{(1)}$ It is doubly degenerate: denote $f_1^{(0)} = \varphi_{10}^{(0)}(xy) = \varphi_1^{(0)}(x)\,\varphi_0^{(0)}(y)$

$$f_2^{(0)} = \varphi_{01}^{(0)}(xy) = \varphi_0^{(0)}(x)\,\varphi_1^{(0)}(y)$$

$n_1=0 \; n_2=1$

$\Rightarrow$ the eigenvalue problem

$$\begin{pmatrix} \langle f_1|H_1|f_1\rangle & \langle f_1|\hat{H}_1|f_2\rangle \\ \langle f_2|H_1|f_1\rangle & \langle f_2|\hat{H}_1|f_2\rangle \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_1^{(1)}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\langle f_1|H_1|f_1\rangle = \alpha\langle\varphi_1|\hat{x}|\varphi_1\rangle\langle\varphi_0|\hat{y}|\varphi_0\rangle = 0 = \langle f_2|H_1|f_2\rangle$$

$$\langle f_1|H_1|f_2\rangle = \alpha\,\langle\varphi_1,\varphi_0|\,\hat{x}\,\hat{y}\,|\varphi_0,\varphi_1\rangle =$$

$$= \alpha\langle 1|\hat{x}|0\rangle\langle 0|\hat{y}|1\rangle = \alpha\frac{\hbar}{2m\omega}\langle 1|\hat{a}+\hat{a}^+|0\rangle\cdot$$

$$\cdot\langle 0|\hat{a}+\hat{a}^+|1\rangle =$$

$$= \frac{\alpha\hbar}{2m\omega}(1)(1); \quad \text{clearly } \langle f_2|\hat{H}_1|f_1\rangle = \frac{\alpha\hbar}{2m\omega}$$

$$\Rightarrow \text{solve} \begin{pmatrix} 0 & \frac{\alpha\hbar}{2m\omega} \\ \frac{\alpha\hbar}{2m\omega} & 0 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_1^{(1)}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\Rightarrow \det\begin{pmatrix} -E_1^{(1)} & \frac{\alpha\hbar}{2m\omega} \\ \frac{\alpha\hbar}{2m\omega} & -E_1^{(1)} \end{pmatrix} = 0 \Rightarrow \left(E_1^{(1)}\right)^2 = \left(\frac{\alpha\hbar}{2m\omega}\right)^2$$

$$\Rightarrow E_1^{(1)} = \pm\frac{\alpha\hbar}{2m\omega}$$

$\Rightarrow$ can find "right" unperturbed combinations now:

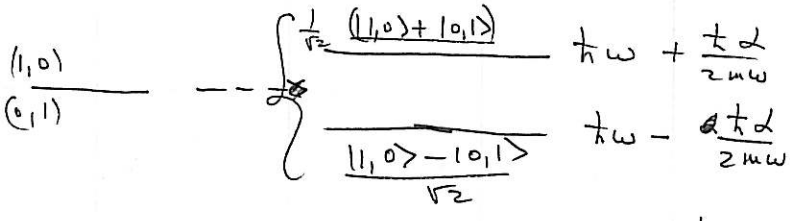
$$\begin{pmatrix} 0 & \frac{\alpha\hbar}{2m\omega} \\ \frac{\alpha\hbar}{2m\omega} & 0 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \pm\frac{\alpha\hbar}{2m\omega}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \pm\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\begin{pmatrix} c_2 \\ c_1 \end{pmatrix} = \pm\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \Rightarrow \begin{cases} c_2 = c_1 = \frac{1}{\sqrt{2}} & \oplus \\ c_1 = -c_2 = \frac{1}{\sqrt{2}} & \ominus \end{cases}$$

$$\Rightarrow |\gamma_1\rangle = \frac{1}{\sqrt{2}}|f_1\rangle + \frac{1}{\sqrt{2}}|f_2\rangle \qquad E^{(1)}_{1+} = +\frac{\alpha\hbar}{2m\omega}$$

$$|\gamma_2\rangle = \frac{1}{\sqrt{2}}|f_1\rangle - \frac{1}{\sqrt{2}}|f_2\rangle \qquad E^{(1)}_{1-} = -\frac{\alpha\hbar}{2m\omega}$$



check that in the "right" basis the standard perturbation theory applies! pages 11, 12, 13

$\Rightarrow$ the degeneracy is splitted!

(3) $\underline{\underline{E_2^{(1)}}}$ Can also find the fate of 2nd excited state, which is 3-fold degenerate! (requires 3x3 matrix problem).

Here we have 3 states:
$$\begin{cases} f_1 = \varphi^{(0)}_{20}(x,y) = \varphi^{(0)}_2(x)\,\varphi^{(0)}_0(y) \\ f_2 = \varphi^{(0)}_{02}(x,y) = \varphi^{(0)}_0(x)\,\varphi^{(0)}_2(y) \\ f_3 = \varphi^{(0)}_{11}(x,y) = \varphi^{(0)}_1(x)\,\varphi^{(0)}_1(y) \end{cases}$$

Work out matrix elements of perturbation $\hat{H}_1$:

clearly $\langle f_i|\hat{H}_1|f_i\rangle = 0$ for $i = 1,2,3$

$$\langle f_1|\hat{H}_1|f_2\rangle = \alpha\,\langle 2_x|\hat{x}|0_x\rangle\langle 0_y|\hat{y}|2_y\rangle =$$

using notation $|n_x\rangle \equiv |\varphi^{(0)}_{n_x}\rangle$

$$= \alpha\left(\frac{\hbar}{2m\omega}\right)\langle 2_x|\hat{a}+\hat{a}^+|0_x\rangle\langle 0_y|\hat{a}+\hat{a}^+|2_y\rangle = 0 \quad \text{because } \langle 0|1\rangle = 0.$$

$$\langle f_1|\hat{H}_1|f_3\rangle = \alpha\,\langle 2_x|\hat{x}|1_x\rangle\langle 0_y|\hat{y}|1_y\rangle = \alpha\frac{\hbar}{2m\omega}\langle 2|\hat{a}+\hat{a}^+|1\rangle\langle 0|\hat{a}+\hat{a}^+|1\rangle$$

$$= \frac{\hbar\alpha}{2m\omega}\sqrt{2}\cdot 1 = \frac{\hbar\alpha}{\sqrt{2}\,m\omega}$$

Since $f_2(x,y) = f_1(y,x)$, we also have

$$\langle f_2 | \hat{H}_1 | f_3 \rangle = \frac{\hbar\alpha}{\sqrt{2}\, m\omega}$$

Since $\hat{H}_1 = \hat{H}_1^+$, also the adjoints are the same:

Indeed $\langle f_3 | \hat{H}_1 | f_1 \rangle = \alpha \frac{\hbar}{2m\omega} \langle 1_x | \hat{a} + \hat{a}^+ | 2_x \rangle \langle 1_y | \hat{a} + \hat{a}^+ | 0_y \rangle =$

$$= \frac{\hbar\alpha}{2m\omega} \sqrt{2} \cdot 1 = \frac{\hbar\alpha}{\sqrt{2}\, m\omega} = \langle f_3 | \hat{H}_1 | f_2 \rangle.$$

Thus the eigenvalue problem reads

$$\begin{pmatrix} 0 & 0 & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} \\ 0 & 0 & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} \\ \frac{\hbar\alpha}{\sqrt{2}\, m\omega} & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = E_2^{(1)} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\Rightarrow \det \begin{pmatrix} -E_2^{(1)} & 0 & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} \\ 0 & -E_2^{(1)} & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} \\ \frac{\hbar\alpha}{\sqrt{2}\, m\omega} & \frac{\hbar\alpha}{\sqrt{2}\, m\omega} & -E_2^{(1)} \end{pmatrix} = 0$$

$$\Rightarrow -E_2^{(1)} \left[\left(E_2^{(1)}\right)^2 - \left(\frac{\hbar\alpha}{\sqrt{2}\, m\omega}\right)^2 \right] + \frac{\hbar\alpha}{\sqrt{2}\, m\omega} \frac{\hbar\alpha}{\sqrt{2}\, m\omega} E_2^{(1)} = 0$$

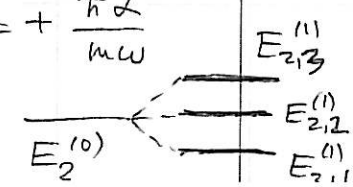
$$\Rightarrow E_2^{(1)} = 0 \quad \text{or} \quad 2\left(\frac{\hbar\alpha}{\sqrt{2}\, m\omega}\right)^2 - \left(E_2^{(1)}\right)^2 = 0$$

$$\Rightarrow E_2^{(1)} = \pm \frac{\hbar\alpha}{m\omega}$$

$\Rightarrow$ three roots (in increasing order):

$$E_{2,1}^{(1)} = -\frac{\hbar\alpha}{m\omega} \; ; \; E_{2,2}^{(1)} = 0 \; ; \; E_{2,3}^{(1)} = +\frac{\hbar\alpha}{m\omega}$$

$\Rightarrow$ the 3-fold degeneracy is lifted completely!



And, for completeness sake, let's find what are the proper linear combinations of unperturbed degenerate states:

$\rightarrow E^{(1)}_{2,1} = -\frac{\hbar\alpha}{m\omega} \Rightarrow$

$$\begin{pmatrix} 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = - \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\begin{cases} \frac{1}{\sqrt{2}} c_3 = -c_1 \\ \frac{1}{\sqrt{2}} c_3 = -c_2 \\ \frac{1}{\sqrt{2}}(c_1 + c_2) = -c_3 \end{cases} \Rightarrow \begin{cases} c_1 = c_2 \\ \sqrt{2}\, c_1 = -c_3 \end{cases}$$

normalization requires: $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$

$$\Rightarrow \quad 2|c_1|^2 + |c_3|^2 = 4|c_1|^2 = 1 \Rightarrow c_1 = \frac{1}{2}$$

$$\Rightarrow |\varphi^{(0)}_{2,1}\rangle = \frac{1}{2}|f_1\rangle + \frac{1}{2}|f_2\rangle - \frac{\sqrt{2}}{2}|f_3\rangle$$

$$= \frac{|f_1\rangle + |f_2\rangle - \sqrt{2}\,|f_3\rangle}{2} \qquad (1')$$

$\rightarrow E^{(1)}_{2,2} = 0 \Rightarrow$

$$\begin{pmatrix} 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} c_3 \\ \frac{1}{\sqrt{2}} c_3 \\ \frac{1}{\sqrt{2}}(c_1 + c_2) \end{pmatrix} = 0 \Rightarrow c_3 = 0$$

$$c_1 = -c_2$$

normalization: $|c_1|^2 + |c_2|^2 + \underbrace{|c_3|^2}_{\to 0} = 1 \Rightarrow 2|c_1|^2 = 1$

$$c_1 = \frac{1}{\sqrt{2}}$$

$$\Rightarrow |\varphi^{(0)}_{2,2}\rangle = \frac{1}{\sqrt{2}}|f_1\rangle - \frac{1}{\sqrt{2}}|f_2\rangle =$$

$$= \frac{|f_1\rangle - |f_2\rangle}{\sqrt{2}} \qquad (2')$$

Finally $E^{(0)}_{2,3} = \frac{\hbar\alpha}{\sqrt{2}\, m\omega}$

$$\Rightarrow \begin{pmatrix} 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}} c_3 \\ \frac{1}{\sqrt{2}} c_3 \\ \frac{1}{\sqrt{2}} (c_1 + c_2) \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \rightarrow \begin{cases} c_1 = c_2 \\ \sqrt{2}\, c_1 = c_3 \end{cases}$$

$\Rightarrow$ normalization: $|c_1|^2 + |c_2|^2 + |c_3|^2 = 2|c_1|^2 + 2|c_1|^2 = 1$

$$\Rightarrow c_1 = \frac{1}{2} \quad \Rightarrow c_3 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow |\varphi^{(0)}_{2,3}\rangle = \frac{1}{2}|f_1\rangle + \frac{1}{2}|f_2\rangle + \frac{\sqrt{2}}{2}|f_3\rangle =$$

$$= \frac{|f_1\rangle + |f_2\rangle + \sqrt{2}\,|f_3\rangle}{2} \quad (3')$$

$(1', 2', 3')$: These are the appropriate linear combinations that evolve, under $\hat{H}_1$, into non-degenerate states with energies $E^{(0)}_2 + E^{(1)}_{2(1,2,3)} = 3\hbar\omega + \begin{cases} -\frac{\hbar\alpha}{m\omega} \\ 0 \\ \frac{\hbar\alpha}{m\omega} \end{cases}$

More on "right" states $|\gamma_1\rangle$ and $|\gamma_2\rangle$.

Found $E_1^{(1)} = \pm \frac{\hbar\alpha}{2m\omega}$ and $|\gamma_{1,2}\rangle = \frac{1}{\sqrt{2}}\left(|f_1\rangle \pm |f_2\rangle\right)$

these are "right" ("good") unperturbed linear combinations.

Calculate $\langle\gamma_1|\hat{H}_1|\gamma_2\rangle = \frac{1}{\sqrt{2}}\left(\langle f_1| + \langle f_2|\right)\hat{H}_1\,\frac{1}{\sqrt{2}}\left(|f_1\rangle - |f_2\rangle\right)$

$$= \frac{1}{2}\left(\underbrace{\langle f_1|\hat{H}_1|f_1\rangle}_{\emptyset} - \langle f_1|\hat{H}_1|f_2\rangle + \langle f_2|\hat{H}_1|f_1\rangle - \underbrace{\langle f_2|\hat{H}_1|f_2\rangle}_{\emptyset}\right)$$

$$= \frac{1}{2}\left(-\frac{\alpha\hbar}{2m\omega} + \frac{\alpha\hbar}{2m\omega}\right) = 0 \quad !$$

see Theorem on pages 259-260 of Griffiths

Thus in this $|\gamma_{1,2}\rangle$ basis we will not encounter any divergencies (in denominator) as the numerator is identically zero!

the symmetry of $\hat{H}_1$ is interchange of x and y axes: $\hat{A}|\gamma_{1,2}\rangle = \pm|\gamma_{1,2}\rangle$

$$\hat{H}_1|\gamma_2\rangle = \frac{\alpha}{\sqrt{2}}\,\hat{x}\,\hat{y}\left(|1_x, 0_y\rangle - |0_x, 1_y\rangle\right) =$$

$$= \frac{\alpha}{\sqrt{2}}\,\frac{\hbar}{2m\omega}\left(\hat{a}_x + \hat{a}_x^+\right)\left(\hat{a}_y + \hat{a}_y^+\right)\left(|1_x\rangle|0_y\rangle - |0_x\rangle|1_y\rangle\right) =$$

$$\left(\hat{a} + \hat{a}^+\right)|1\rangle = |0\rangle + \sqrt{2}\,|2\rangle$$

$$\left(\hat{a} + \hat{a}^+\right)|0\rangle = |1\rangle$$

$$= \frac{\alpha}{\sqrt{2}}\,\frac{\hbar}{2m\omega}\left[\left(|0_x\rangle + \sqrt{2}\,|2_x\rangle\right)|1_y\rangle - |1_x\rangle\left(|0_y\rangle + \sqrt{2}\,|2_y\rangle\right)\right]$$

$$= \frac{\alpha}{\sqrt{2}}\,\frac{\hbar}{2m\omega}\left[\underbrace{\left(|0_x\rangle|1_y\rangle - |1_x\rangle|0_y\rangle\right)}_{-\sqrt{2}|\gamma_2\rangle} + \sqrt{2}\left(|2_x\rangle|1_y\rangle - |1_x\rangle|2_y\rangle\right)\right]$$

$$= -\frac{\hbar\alpha}{2m\omega}|\gamma_2\rangle + \frac{\hbar\alpha}{2m\omega}\left(|2_x, 1_y\rangle - |1_x, 2_y\rangle\right)$$

this is state of high energy: $N=3$ $\quad n_x = 2 \quad n_y = 1 \Rightarrow E = \hbar\omega(3+1) = 4\hbar\omega$

Let us denote this state $|f_4\rangle = \frac{|2_x, 1_y\rangle - |1_x, 2_y\rangle}{\sqrt{2}}$

$$\langle f_4 | f_4 \rangle = 1$$

Then $\hat{H}_1 |\gamma_2\rangle = -\frac{\hbar\alpha}{2m\omega} |\gamma_2\rangle + \frac{\sqrt{2}\hbar\alpha}{2m\omega} |f_4\rangle$

Similarly, consider what obtains of $|\gamma_1\rangle$:

$$\hat{H}_1 |\gamma_1\rangle = \frac{\alpha}{\sqrt{2}} (\hat{a}_x + \hat{a}_x^+)(\hat{a}_y + \hat{a}_y^+) \frac{\hbar}{2m\omega} \left(|1_x, 0_y\rangle + |0_x, 1_y\rangle\right)$$

$$= \frac{\alpha}{\sqrt{2}} \frac{\hbar}{2m\omega} \left[\left(|0_x\rangle + \sqrt{2}|2_x\rangle\right)|1_y\rangle + |1_x\rangle\left(|0_y\rangle + \sqrt{2}|2_y\rangle\right) \right]$$

$$= \frac{\alpha}{\sqrt{2}} \frac{\hbar}{2m\omega} \left[\underbrace{\left(|0_x\rangle|1_y\rangle + |1_x\rangle|0_y\rangle\right)}_{\sqrt{2}|\gamma_1\rangle} + \sqrt{2}\left(|2_x\rangle|1_y\rangle + |1_x\rangle|2_y\rangle\right) \right]$$

$$= \frac{\hbar\alpha}{2m\omega} |\gamma_1\rangle + \frac{\hbar\alpha}{\sqrt{2}m\omega} \underbrace{\frac{|2_x, 1_y\rangle + |1_x, 2_y\rangle}{\sqrt{2}}}_{|f_5\rangle}$$

Thus $\hat{H}_0 |f_5\rangle = \hbar\omega(2+1+1)|f_5\rangle = 4\hbar\omega |f_5\rangle$

$\Rightarrow$ $|f_5\rangle$ is the state of the same energy ($4\hbar\omega$) as $|f_4\rangle$ is, but they are orthogonal:

$$\langle f_4 | f_5 \rangle = \frac{1}{\sqrt{2}} \left(\langle 2_x, 1_y| - \langle 1_x, 2_y|\right) \frac{1}{\sqrt{2}} \left(|2_x, 1_y\rangle + |1_x, 2_y\rangle\right)$$

$$= \frac{1}{2} \left(\langle 2_x, 1_y | 2_x, 1_y\rangle - \langle 1_x, 2_y | 1_x, 2_y\rangle\right) = \frac{1-1}{2} = 0!$$

Finally, ~~using~~ this we can calculate the second order corrections to $|\gamma_{1,2}\rangle$:

for γ_1

$$E^{(2)}_{1(\gamma_1)} = \sum_{k\neq\gamma_1}{}' \frac{|\langle\varphi_k^{(0)}|\hat{H}_1|\gamma_1\rangle|^2}{E^{(0)}_{\gamma_1} - E^{(0)}_k} = \left(\begin{array}{c}\text{the only choice}\\ \varphi_k^{(0)} = f_5\end{array}\right) =$$

$$= \frac{|\langle f_5|\hat{H}_1|\gamma_1\rangle|^2}{E_{\gamma_1} - E_{f_5}} = \frac{\left(\frac{\hbar\alpha}{\sqrt{2}m\omega}\right)^2}{2\hbar\omega - 4\hbar\omega} = -\frac{\alpha^2\hbar}{4m^2\omega^3}$$

for γ_2

$$E^{(2)}_{1(\gamma_2)} = \sum_{k\neq\gamma_2}{}' \frac{|\langle\varphi_k^{(0)}|\hat{H}_1|\gamma_2\rangle|^2}{E_{\gamma_2} - E^{(0)}_k} = \left(\begin{array}{c}\text{here obviously}\\ \varphi_k^{(0)} = f_4\end{array}\right)$$

$$= \frac{|\langle f_4|\hat{H}_1|\gamma_1\rangle|^2}{2\hbar\omega - 4\hbar\omega} = -\frac{\alpha^2\hbar}{4m^2\omega^3}$$

Thus, to this order, $E_{\gamma_1} = \underbrace{2\hbar\omega}_{E^{(0)}_1} + \underbrace{\frac{\hbar\alpha}{2m\omega}}_{E^{(1)}_{1+}} - \underbrace{\frac{\hbar\alpha^2}{4m^2\omega^3}}_{E^{(2)}_{1(\gamma_1)}}$

$$E_{\gamma_2} = 2\hbar\omega + \frac{\hbar\alpha}{2m\omega} - \frac{\hbar\alpha^2}{4m^2\omega^3}$$

And correction to $|\gamma_{1,2}\rangle$ themselves:

$$|\tilde{\gamma}_1\rangle = |\gamma_1^{(0)}\rangle + \sum_{k\neq\gamma_1} |\varphi_k^{(0)}\rangle \frac{\langle\varphi_k^{(0)}|\hat{H}_1|\gamma_1\rangle}{E_{\gamma_1} - E^{(0)}_k} = \left(\begin{array}{c}\text{see } E^{(2)}_{1(\gamma_1)}\\ \text{above}\end{array}\right) =$$

$$= |\gamma_1^{(0)}\rangle + |f_5\rangle \frac{\frac{\hbar\alpha}{\sqrt{2}m\omega}}{-2\hbar\omega} = |\gamma_1^{(0)}\rangle - \frac{\alpha}{2^{3/2}m\omega^2}|f_5\rangle$$

And $|\tilde{\gamma}_2\rangle = |\gamma_2^{(0)}\rangle - \frac{\alpha}{2^{3/2}m\omega^2}|f_4\rangle$